# SIMATS ENGINEERING

## Department of Computer Science & Engineering

### ASSESSMENT TOOL 1- CONSTRAINT-BASED PROBLEM SOLVING

| | | | |
|---|---|---|---|
| **Course Code:** | CSA12 | **Course Title:** | Computer Architecture |
| **CO Assessed:** | CO5 | **Assessment:** | ASSESSMENT TOOL1- Constraint-Based Problem Solving |
| **Weightage:** | 40% | **Total Marks:** | 25 |
| **CO5 Statement:** | *Construct I/O communication mechanisms by comparing memory-mapped and I/O-mapped techniques, interrupt handling (vectored, hardware, software), DMA controllers, and advanced bus interfaces including PCIe, USB, NVMe, and Thunderbolt.* | | |
| **Student Name:** | K. Yoganand | **Reg.No.:** | 192525173 |
| **Department:** | CSE (AI/ML) | **Date:** | |

83%

**ANNEXURE A**

**Constraint-Based Problem Solving**

1.Construct an I/O communication mechanism for a real-time embedded system with the following constraints:

- CPU utilization must be below 15%
- Multiple sensors generate interrupts every 1 ms
- Use vectored interrupt handling
- Select between memory-mapped and I/O-mapped I/O for optimal latency

2.Design an I/O subsystem under these constraints:

- High-speed storage requires throughput > 3 GB/s
- Must use NVMe over PCIe
- CPU should not handle bulk data transfer
- Integrate DMA controller with interrupt notification

3.Construct a multi-device communication architecture with constraints:

- 6 peripherals using USB, 2 high-speed devices using Thunderbolt
- Avoid interrupt starvation
- Use hardware and software interrupts
- Ensure fair bus arbitration

4.Design a data acquisition system with constraints:

- Continuous data stream at 500 MB/s
- Minimal CPU intervention required
- Must compare DMA transfer vs interrupt-driven I/O
- Use memory-mapped I/O for device registers

5.Construct an interrupt handling mechanism with constraints:

- Support nested interrupts
- Maximum interrupt latency < 10 µs
- Use vectored interrupts for high-priority devices
- Use software interrupts for background tasks

**Code of Conduct:**

I K. Yoganand (192525173) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

**Signature of the student:** K. Yoganand

MARKS ALLOCATION

| | Problem Understanding (25%) | Constraint Analysis (25%) | Solution Development (25%) | Application of Concepts (15%) | Justification (10%) |
|---|---|---|---|---|---|
| Q1 | 2 | 2 | 2 | 1.5 | 1 |
| Q2 | 2 | 2 | 2 | 1.5 | 1 |
| Q3 | 2.5 | 2.5 | 1.5 | 1 | 0.5 |
| Q4 | 2.5 | 2 | 1.5 | 1 | 0.5 |
| Q5 | 2 | 2.5 | 2 | 1.5 | 1 |

41.5

**RUBRICS FOR CALCULATION:**

| Criteria | Weightage | Level 4 – Exemplary | Level 3 – Proficient | Level 2 – Developing | Level 1 – Beginning |
|---|---|---|---|---|---|
| Problem Understanding | 25% 2.5 | Clearly defines the problem and identifies all relevant constraints accurately | Identifies most constraints with minor gaps | Partial understanding; misses key constraints | Fails to identify constraints correctly |
| Constraint Analysis | 25% 2.5 | Analyzes constraints effectively and prioritizes them logically | Provides reasonable analysis with some prioritization | Limited analysis; weak prioritization | No meaningful analysis or prioritization |
| Solution Development | 25% 2.5 | Develops optimal and feasible solutions satisfying all constraints | Develops workable solutions with minor limitations | Solutions partially satisfy constraints | Solutions do not meet constraints |
| Application of Concepts | 15% 1.5 | Effectively applies appropriate concepts, methods, or tools | Applies concepts with minor errors | Limited application; requires guidance | Unable to apply relevant concepts |
| Justification | 10% 1 | Provides strong justification for decisions with logical reasoning | Provides justification with some clarity | Weak or unclear justification | No justification provided |